Exploring DeepFakes: Face Reenactment

Comparative Architecture Analysis: Autoencoders, Keypoint-Based FOMM & Rectified Flow

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Deep Learning & Computer Vision

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Outline

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3. Method 2: Keypoint-Based FOMM
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Problem Formulation: Face Reenactment

Core Objective

- Every method works with two roles.
- The **identity image** supplies the face.
- The **motion source** supplies the movement.
- Each method uses the pair in its own way.

Key Technical Challenges

- **Identity preservation:** Avoid warping face features into driving face.
- **Non-rigid deformation:** Facial skin stretches (not just rigid 2D shift).
- **Occlusion handling:** Mouth opening reveals unseen areas (teeth, tongue).

Input / Output Setup

Inputs:

- Identity image: supplies the face.
- Motion source: supplies the movement.

Each method gives the inputs their own names. **Output:**

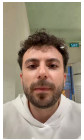
- Reenacted video of the identity face
- Mimicking expressions and head pose of the motion source

Dataset Preparation & Autoencoder Architecture

- **Goal & Motivation:** Swap faces between Person A and Person B while keeping motion. Built custom autoencoder to experience basics before pretrained tools.
- **Preprocessing:** Recorded short videos, extracted every 5th frame. Haar Cascade detection with **+25% margin**. Resized to 128×128 (0–1 scaled).
- **Architecture:** 1 shared Encoder (4 conv layers) and 2 Decoders (4 transposed-conv layers each, Sigmoid activation).
- **Face Swap Scheme:** Pass Person A through Decoder B, and Person B through Decoder A.

Subject	Prepared Faces	Resolution
Person A	100	128×128
Person B	138	128×128
Total	**238**	**128×128**

Input Videos (First Frames)



Person A (100 imgs)
[person_A.mp4]



Person B (138 imgs)
[person_B.mp4]

Custom Autoencoder: Training Loss & Video Outputs

Training Schedule (150 Epochs)

- First trained 50 epochs, saved weights.
- Resumed for another 100 epochs (150 total).

Epoch	Person A MSE	Person B MSE
146	0.003046	0.001939
147	0.003013	0.001929
148	0.003014	0.001905
149	0.002996	0.001882
150	**0.002935**	**0.001866**

Both models converged steadily; Person B achieved lower training loss.

Generated Video Outputs

Epochs	A → B Video	B → A Video
50 ep	`deepfake_A_to_B`	`deepfake_B_to_A`
150 ep	`deepfake_150_A_to_B`	`deepfake_150_B_to_A`



Comparison A → B (Input | 50ep | 150ep)

[deepfake_150_A_to_B.mp4] [comparison_A_to_B.mp4]

Custom Autoencoder: Observations & Failure Analysis

Observations & Hypotheses

- **Observation.** The swap failed in this run: the A $\rightarrow$ B face did not look like Person B, and B $\rightarrow$ A did not look like Person A.
- **Observation.** The 50 and 150 epoch videos showed no clear improvement.
- **Hypothesis.** Pixel-wise MSE may train the model to recreate the original face, not to swap it.
- **Hypothesis.** 238 training images may be too few.
- **Hypothesis.** The shared latent code may entangle identity with pose and expression.
- **Status.** We did not test these causes.

Triplet Video Comparison Frame



Comparison B $\rightarrow$ A (Input | 50ep | 150ep)

[comparison_B_to_A.mp4]

Key Takeaway

Possible cause: the shared latent code entangled identity with facial pose and expression. This was not tested separately.

Benchmarking Pretrained Models (FaceFusion)

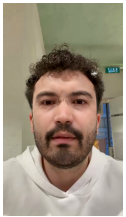
Inference Setup

- Tested 3 pretrained models in FaceFusion (no retraining).
- Source face gives identity; target video gives movement.

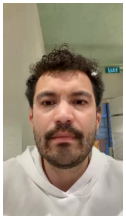
Model	Output Size	Frames	Duration
HyperSwap 1A	576×1024	576	$\sim$19.25 s
InSwapper 128	576×1024	576	$\sim$19.25 s
SimSwap 256	576×1024	576	$\sim$19.25 s

Visual Comparison

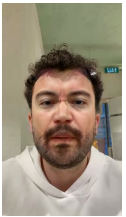
- **HyperSwap & InSwapper:** Fairly natural and consistent.
- **SimSwap:** Followed motion, but showed artifacts around eyes and skin texture.



HyperSwap 1A
[hyperswap_1a_256.mp4]



InSwapper 128
[inswapper_128.mp4]



SimSwap 256
[simswap_256.mp4]

First frames: Erdem on Yusuf swaps

Autoencoders: Evaluation & Lessons Learned

Overall Evaluation

- Training our own autoencoder was not successful for face swapping.
- FaceFusion pretrained models gave much more promising results.
- Showed the importance of using models designed specifically for face swapping.
- Having only 100 images for Person A and 138 for Person B was a major limitation.

Lessons Learned (Erdem's Takeaways)

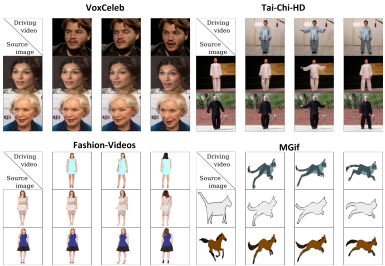
- **Low loss $\neq$ usable result:** The model got better at reconstructing faces, but not at swapping them. Another 100 epochs did not fix this.
- **Beyond running tools:** Building our own model helped us understand preprocessing, training, and evaluation hands-on.
- Pretrained models helped us compare how task-specific designs improve the result.

First Order Motion Model (Siarohin et al., NeurIPS 2019)

Core Innovations

- **No Pretrained Landmarks:** Doesn't need 68 landmark detectors or 3D models (3DMM).
- **Self-Supervised:** Discovers keypoints automatically from raw video pairs.
- **Decoupled Repr.:** Appearance from source S; motion from driving D.
- **Domain Agnostic:** Works on human faces, bodies (Tai-Chi), and animals.

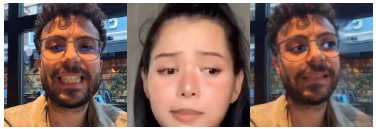
No adversarial loss: this project trains FOMM without a discriminator.



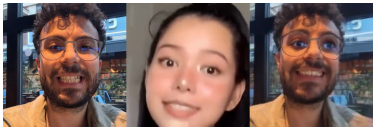
FOMM teaser: VoxCeleb, Tai-Chi, Fashion (Siarohin et al., 2019).

Experimental Results: Bella Poarch Benchmark

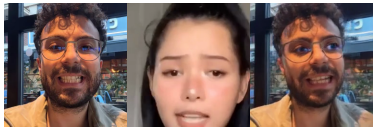
Triplets: [Source Portrait | Driving Frame | Synthesized Result]



Frame 30 (1.0s): Head Tilt



Frame 286 (9.5s): Smile & Squint



Frame 20 (0.7s): Mouth Opening

Demo Video Link

- **3×3 Multi-Subject Comparison Video (All Members):** Watch Online (External Link)

Failure Mode Experiment: Dental Braces

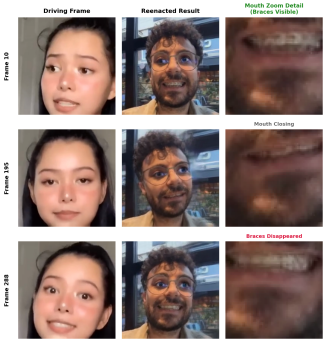
Experimental Setup

- **Goal:** Test FOMM on rare features absent from training data (OOD).
- **Subject:** Source portrait with metallic dental braces.
- **Motion:** Sequence from head tilt to lip closure and wide smile.

Hypothesis & Expected Regimes

- VoxCeleb training data has no faces with braces.
- **Warping ($O \approx 1$):** Samples source pixels $\rightarrow$ braces preserved.
- **Inpainting ($O \approx 0$):** Large deformation masks mouth $\rightarrow$ model synthesizes generic teeth.
- **Prediction:** Inpainting erases braces using dataset average.

Failure Mode: Vanishing Dental Braces



Frame 10 (top), Frame 195 (middle), Frame 288 (bottom).

Observed Generation Regimes:

- **Frame 10 (0.3s) — Warping ($O \approx 1$):** Small shift preserves flow; metallic brackets transfer cleanly.
- **Frame 195 (6.5s) — Lip Closure:** Lips close, naturally hiding mouth interior.
- **Frame 288 (9.6s) — Inpainting ($O \approx 0$):** Extreme smile triggers inpainting; generator paints generic teeth, erasing braces.

O values: visual reading, not a stored measurement.

Technical Mechanism: Feature Warping & Occlusion

Synthesis Formulation

$$\text{Output} = D(W(E(S),\ T) \odot O)$$

One decoder path: no inpainting branch, no RGB blend.

Mechanism Steps

- **Encode:** the source S becomes features $E(S)$.
- **Warp:** the dense flow T moves the features.
- **Weight:** the mask O multiplies the warped features.
- **Decode:** $D(\cdot)$ renders the whole output frame.

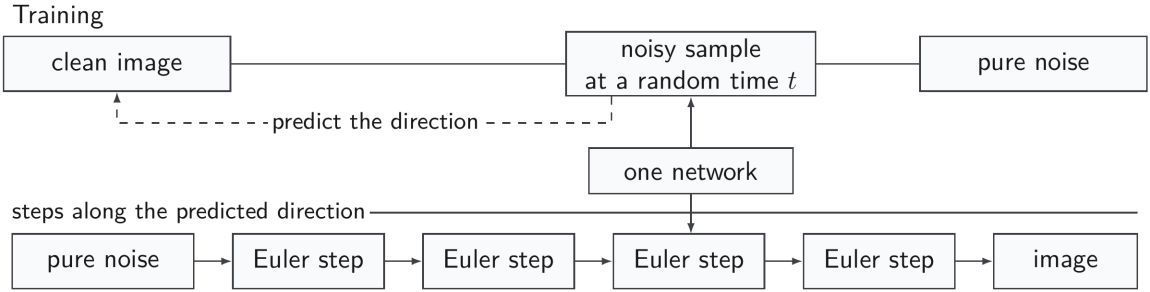
Core Finding

- At the mouth, the occlusion mask fails ($O \approx 0$).
- $O \approx 1$ regions keep the warped source details.
- $O \approx 0$ regions fall back on learned appearance priors.

Failure Mode Demo Video

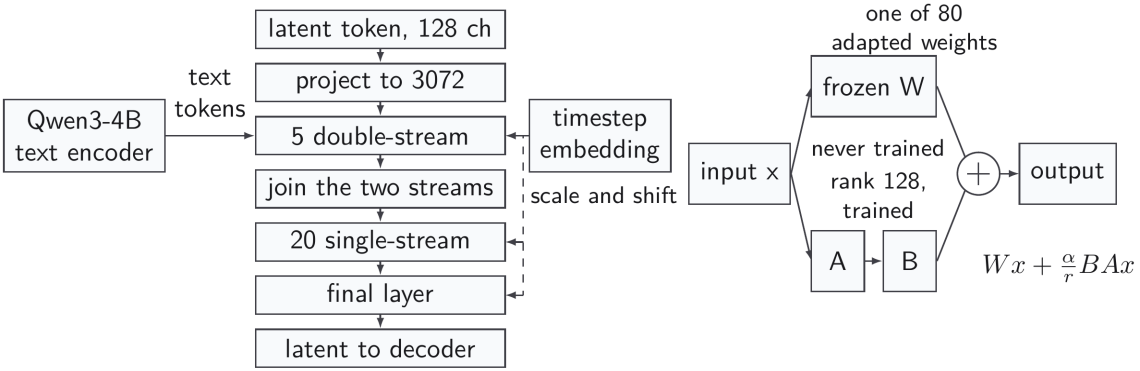
- **Dental Braces Video:**
  Watch Online (External Link)

Rectified Flow Fundamentals



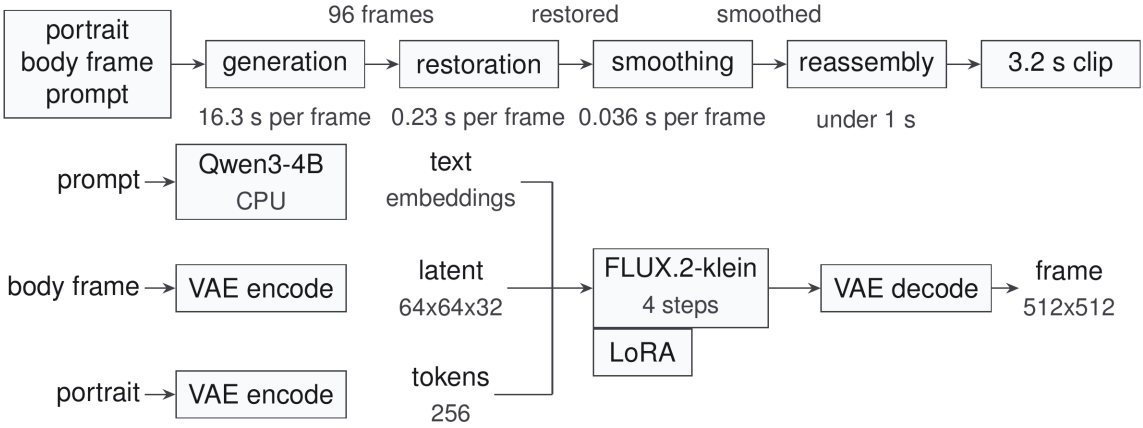
- **Training.** The image and noise mix on one straight line at a random time.
- **Generation.** The model predicts the direction, and the sampler steps along it.
- **Our setup.** A released checkpoint, already step distilled, so four steps are enough.

Architecture: FLUX.2-klein and LoRA



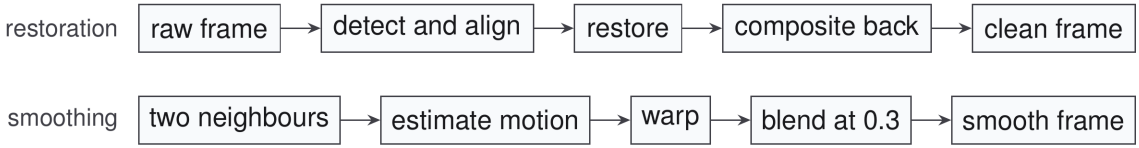
- **Flow model.** A 4 billion parameter rectified flow transformer.
- **Layers.** Five double-stream blocks, then 20 single-stream blocks.
- **Adapter.** It adapts 80 weights, and the base weights stay frozen.

The Generation Pipeline on 4 GB



- **Why it fits.** The text encoder runs once on the CPU, then its memory is freed.
- **Safety gate.** A frame with more than one face is refused, and there is no bypass.

Cleanup Stages and Results



Stage	Face change	p95	Consecutive identity	Identity to portrait	Hair noise
Control	6.657	12.3488	0.985	−0.036	31.600
Raw swap	7.451	17.3794	0.923	0.215	32.570
Restoration	7.160	16.2097	0.927	0.215	32.072
Flow on raw	6.666	15.6581	0.939	0.215	32.060
Restoration + flow	6.630	15.2802	0.938	0.215	31.731
Deflicker	5.627	12.6650	0.936	0.211	31.896

Settings: strength 0.6 for flow on raw, 0.3 for restoration + flow.

Reading: these metrics are temporal change indicators, not proof; motion, lighting, texture change and flicker all enter the number, so read them together with the visual observation.

- **Recommended.** Restoration, then optical flow at strength 0.3.

Reassembly and the Boil



Raw assembly



Flow on the raw frames



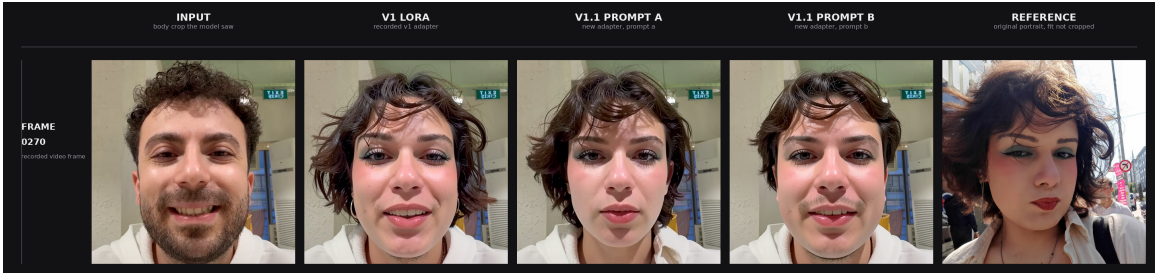
Restoration + flow

- **Reassembly.** H.264 at 29.969 fps, with the original audio muxed back.
- **Watch the boil.** It is clear in the first clip and reduced in the last one. Some frames still jump.

Adapter Versions and Failure Analysis

	v1	v1.1
Seconds per frame	16.34	about 199
Peak VRAM	3.60 GiB	**4.68 GiB**
Adapter rank	128	**512**
96 frames	26 min	about 5.3 h

- **Measured.** Only v1 ran the metric script.
- **Memory.** v1.1 went past the card, 12x slower.
- **Judgement.** We preferred v1 by eye, so the clip kept v1.
- **Tail.** The worst frames still jump.



Five columns: input body crop, v1, v1.1 with prompt A, v1.1 with prompt B, and the reference portrait.

Comparative Analysis

	Autoencoder	**FOMM (Keypoints)**	**Rectified Flow**
What it does	Shared encoder and two decoders; face swap by crossing decoders.	Animates a source photo with 10 unsupervised keypoints and Jacobians.	Head swap with FLUX.2-klein-4B and a LoRA adapter, four steps.
Input	238 prepared face images: 100 Person A, 138 Person B, 128 $\times$ 128.	Source portrait 512 $\times$ 512; driving clip 256 $\times$ 256, 320 frames.	Body crop 512 $\times$ 512, selfie portrait 256 $\times$ 256, fixed prompt.
Project result	Low MSE, but no identity swap; pretrained FaceFusion looked better.	256 $\times$ 256 clip in about 2.5 min; rare braces vanish.	96-frame 512 $\times$ 512 clip with audio; identity stays positive.
Main limitation	MSE reconstructs the input; it does not measure swap quality.	Inpainting follows the training average; no memory across frames.	Per-frame generation boils; the tail still jumps.

Ethical Implications & DeepFake Detection

Societal Risks

- **Impersonation:** Reenacting individuals without their consent.
- **Misinformation:** Generating fabricated videos to deceive the public.

Detection Strategies

- **Blinking and rPPG:** Detection cues reported in the literature; this project ran no such test.
- **Frequency artifacts:** Reported in the literature; neural upsampling leaves high-frequency noise.
- **Watermarking:** A separate measure, independent of C2PA.
- **C2PA:** A signed provenance and edit-history manifest, not only a capture-time stamp. It is not a deepfake detector by itself.

Conclusion

Summary

- **Autoencoder:** Low loss, but no identity swap; pretrained models gave better results.
- **FOMM:** Head turns and blinks transferred smoothly; inpainting erased rare details (braces).
- **Rectified Flow:** Head swap ran on a 4 GB GPU; the worst frames still jump.
- **Lesson:** A low loss does not mean a usable swap; use a model built for the task.

Project Repository

Code, inputs, and outputs:

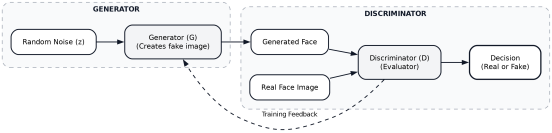
`https://github.com/yusufarsln98/face-reenactment`

Questions & Discussion

Standard GAN Architecture

How GANs Work (Goodfellow et al., 2014)

- Two neural networks competing in an adversarial setup:
- **Generator (G):** Takes noise **z** and learns to generate realistic face images.
- **Discriminator (D):** Evaluates images and decides if real or fake.
- **Adversarial training:** G gets feedback from D and learns photorealistic textures.



The Controllability Issue in Vanilla GANs

Why Unconditional GANs Fall Short

- Generates arbitrary faces from noise $\mathbf{z}$.
- Identity, pose, and emotion are entangled in $\mathbf{z}$.
- Cannot say: *"keep Person A, copy Person B's expression"*.

What Face Reenactment Requires

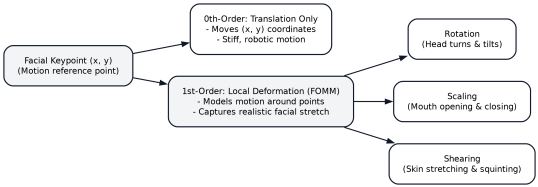
- Keep source identity fixed: appearance from S.
- Motion driven externally: dynamics from D.



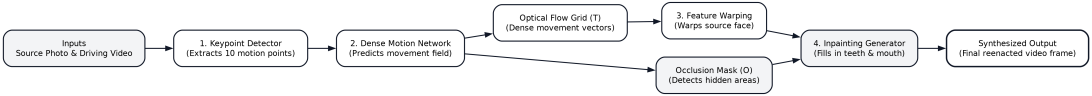
Motion Representation: 0th-Order vs. 1st-Order Motion

How Motion is Represented

- **0th-Order (Translation Only):**
  - Moves points in 2D space (x, y).
  - Cannot stretch skin; movement looks stiff and robotic.
- **1st-Order (FOMM / Local Affine):**
  - Estimates a Jacobian matrix around each point.
  - Captures rotation, scale (open mouth), and shear (skin stretch).



FOMM Architectural Pipeline



1. Keypoint Detector

Hourglass network extracts $K = 10$ unsupervised keypoints and Jacobians.

2. Dense Motion Network

Combines sparse keypoint motions into dense optical flow (T) and mask (O).

3. Generator (Warp, Weight, Decode)

Warps source features with flow T, weights them by the mask O, then decodes. Occluded areas ($O \approx 0$) fall back on learned priors.

Inference Pipeline: Python Pseudocode

Frame-by-Frame Execution Loop

```
# Reference keypoints: source and first driving frame
kp_source = keypoint_detector(source_image)
kp_init = keypoint_detector(driving_video[0])

for frame_d in driving_video:
    kp_driving = keypoint_detector(frame_d)
    # Normalise kp and Jacobians to frame 0
    kp_driving = normalize_kp(kp_source,
        kp_driving, kp_init)
    # Dense motion also needs the source image
    flow, occ = dense_motion_net(source_image,
        kp_source, kp_driving)
    # Warp features, apply the mask, decode
    out_frame = generator(source_image, flow, occ)

    yield out_frame
```

Core Modules

- `keypoint_detector`: Extracts 10 unsupervised keypoints.
- `dense_motion_net`: Estimates flow and mask from the source image and keypoints.
- `generator`: Warps source features, applies the mask, then decodes.

Architectural Properties

No Recurrence: The loop carries no hidden state across frames ($t \to t+1$).
No Adversarial Loss: This configuration trains without a discriminator.

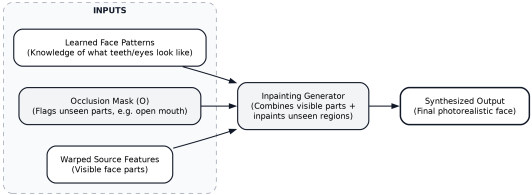
Self-Occlusion & Inpainting Mechanism

The Occlusion Problem

- Turning head or opening mouth reveals unseen areas in S (teeth, tongue).
- Pure warping has no source pixels $\rightarrow$ causes stretching or black holes.

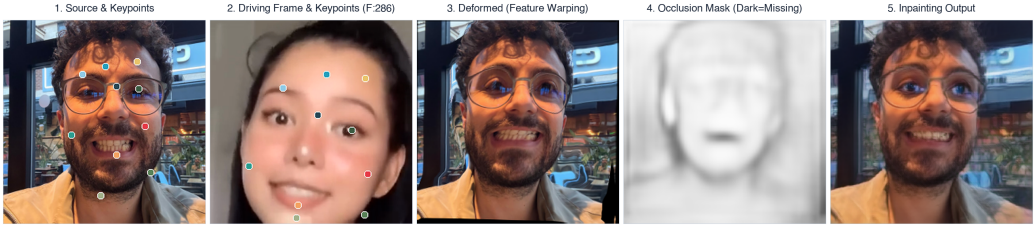
The FOMM Solution

- **Occlusion Mask (O):** Flags missing pixels ($O \approx 0$).
- **Inpainting:** Generator fills missing pixels using learned facial priors.



Pipeline Verification: Intermediate Stages

5 Key Stages Extracted from Our Local Inference Run



1. Source Points
S reference keypoints

2. Driving Points
D tracking points

3. Warped Source (RGB)
Flow-deformed source image

4. Occlusion Mask
Dark = occluded regions

5. Inpainted Result
Final synthesized frame